

This handout includes space for every question that requires a written response. Please feel free to use it to handwrite your solutions (legibly, please). If you choose to typeset your solutions, the `README.md` for this assignment includes instructions to regenerate this handout with your typeset $\text{\LaTeX}$ solutions.

1.a

$\text{Rain} \rightarrow \text{WetGrass}$ and $\text{Sprinkler} \rightarrow \text{WetGrass}$. There is no edge between Rain and Sprinkler, since neither one causes the other.

1.b

The joint is the product of each variable's local conditional probability given its parents:

$$\mathbb{P}(\text{Rain} = r, \text{Sprinkler} = s, \text{WetGrass} = w) = p(r) p(s) p(w \mid r, s).$$

1.c

Sum the joint over both values of Rain and both values of Sprinkler:

$$\mathbb{P}(\text{WetGrass} = w) = \sum_r \sum_s p(r) p(s) p(w \mid r, s).$$

1.d

The grass is wet with probability 0.9 when it rained and the sprinklers were on, so it stays dry with probability $1 - 0.9 = 0.1$:

$$\begin{aligned} \mathbb{P}(\text{Rain} = 1, \text{Sprinkler} = 1, \text{WetGrass} = 0) &= p(\text{Rain} = 1) p(\text{Sprinkler} = 1) p(\text{WetGrass} = 0 \mid \text{Rain} = 1, \text{Sprinkler} = 1) \\ &= 0.2 \times 0.5 \times 0.1 = 0.01. \end{aligned}$$

1.e

- (a) True.
- (b) False.
- (c) True.

2.d

2.g

3.a

3.b

3.c

3.g

4.d

4.e

5.a

5.b

5.c

5.d

6.a

6.b